Deployment of Genome Databases for Algae Using Galaxy Genome Annotation

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http://abims.sb-roscoff.fr/resources/genomic_resources - support.abims@sb-roscoff.fr

Genomic context

+50 brown algae sequenced

- Data :
  - Genomes
  - Gene prediction
  - Functional annotations
  - RNAseq

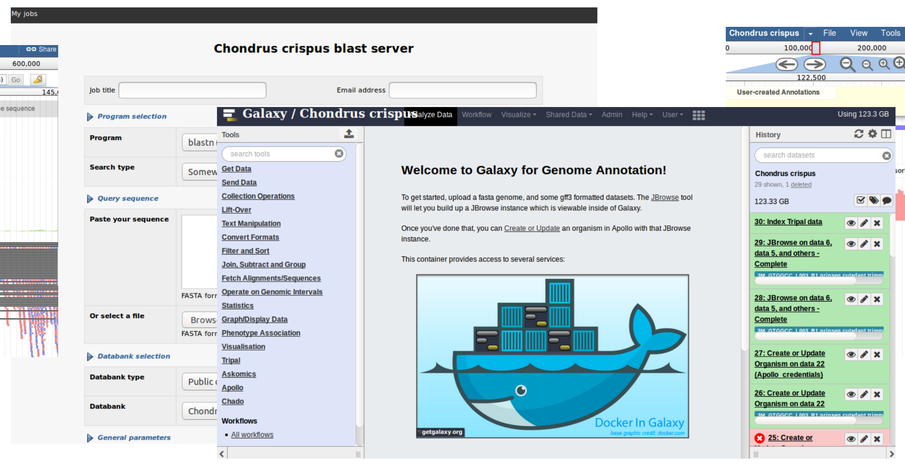
Needs of biologists:
- ✓ Store, centralize
- ✓ Visualize
- ✓ Explore, analyze
- ✓ Edit, add data
- ✓ Share, export, publish

A set of tools

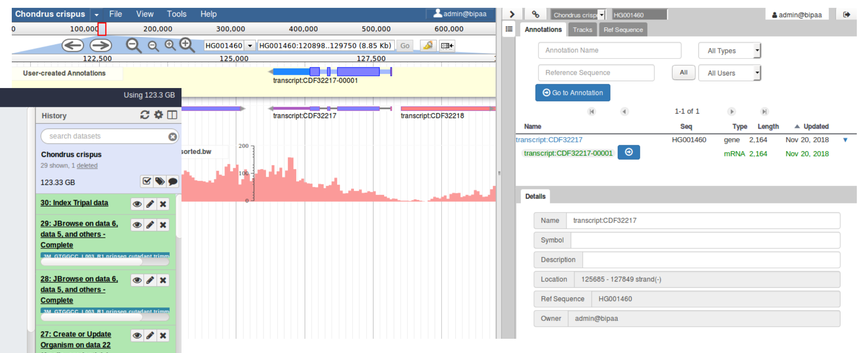
For each genome, a set of web applications:
- **Chado**: database
- **JBrowse**: genome browser
- **Apollo**: collaborative manual curation of genome annotation
- **Tripal**: gene reports with functional annotation
- **Search engine**: indexing data with Elasticsearch
- **Blast server**: simple web interface to launch blast jobs
- **Wiki**: coordination of manual curation

Data visualization

Data analysis



Data contribution



Deployment with Galaxy Genome Annotation

Manual deployment of all applications is not reasonable as the number of brown algae genomes resources to integrate is growing: error-prone, time-consuming, cost of maintenance in operational condition.

New deployment environment:

Galaxy Genome Annotation

https://galaxy-genome-annotation.github.io

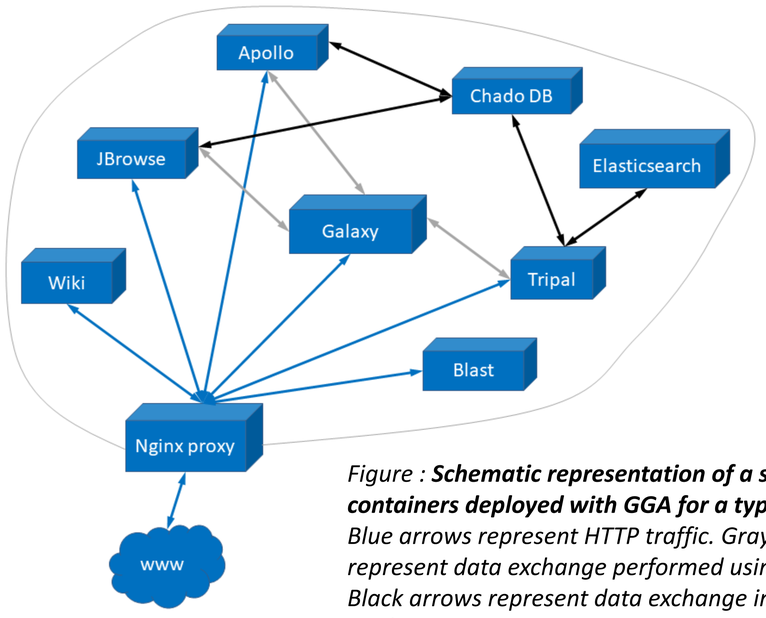


*Figure : **Schematic representation of a set of Docker containers deployed with GGA for a typical genome.** Blue arrows represent HTTP traffic. Gray arrows represent data exchange performed using Galaxy tools. Black arrows represent data exchange inherent in applications.*

The Galaxy Genome Annotation project

A tool suite mainly based on the GMOD tools and the Galaxy web portal:
- Dockerized GMOD tools (JBrowse, Apollo, Tripal, Chado)
- Python libraries to make them work together
- Galaxy tools to interact with GMOD applications

Open-source licenses. Initiative from Galaxy and GMOD communities.

See poster #37A: *"GGA: Easier Genome Annotation using Galaxy and GMOD Tools"*

Implementation

1 genome = 1 set of reusable dockerized web applications

Galaxy as data loading orchestrator for administrators: tools / workflows to load data into JBrowse, Apollo, Tripal, Chado

Benefits

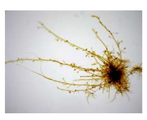
Automatic deployment
Human intervention reduced to the minimum
Reproducibility
1 genome = docker-compose.yml + raw data (fasta, gff, bam) + Galaxy workflow
Flexibility
Customizable set of applications for each genome, all configured in a single file
Scalability
Possibility to dispatch Docker containers on a cluster of servers (e.g. Rancher)

GGA on ABiMS

Implementation on the ABiMS platform

Before 2018:
11 individual genomes browsers (JBrowse) + 6 for curation (Apollo) + Chado database
From 2018:
Implementation of the GGA suite, deployed on 1 server (40 CPUs, 125Gb RAM):
- Red alga *Chondrus crispus*: 1 genome + gene prediction + functional annotations + RNAseq *(Thanks to Jonas Collén[1] for the data)*
- Brown alga *Ectocarpus siliculosus*: 4 genomes + gene prediction

Chondrus crispus


Ectocarpus siliculosus

Future developments

Deployment of the +50 brown algae genomes (roadmap: *E. siliculosus*, *E. subulatus*, *S. japonica*, *C. okamuranus*, *L. elsbetiae*)
Strengthening the infrastructure (cluster coupling)
Web portal as entry point for the genome databases
Support for more types of data (orthology, expression, genotype, phenotype, etc.)
Coupling with genome-scale metabolic networks (http://aureme.genouest.org)
Continue contributing to the evolution of the GGA code